Math 550, S20 Name: Answer

Quiz 1

<u>Instructions:</u> **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.**

Problem 1. (6 points) Given two vectors $\mathbf{u} = 3\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}$ and $\mathbf{v} = (4, 0, 3)$ in $\mathbb{R}^3$.

(a) Find the angle between the two vectors $\mathbf{u}$ and $\mathbf{v}$.

$$\|u\| = \sqrt{3^2+(-2)^2+6^2} = \sqrt{49} = 7$$

$$\|v\| = \sqrt{4^2+0^2+3^2} = \sqrt{25} = 5$$

$$u \cdot v = (3,-2,6)\cdot(4,0,3) = 3\cdot 4 + (-2)\cdot 0 + 6\cdot 3 = 30$$

$$\cos(\theta) = \frac{u\cdot v}{\|u\|\|v\|} = \frac{30}{7\cdot 5} = \frac{6}{7} \Rightarrow \theta = \cos^{-1}\left(\frac{6}{7}\right)$$

(b) Find the vector projection of $\mathbf{v}$ onto $\mathbf{u}$, $\text{proj}_{\mathbf{u}}\mathbf{v}$.

$$\text{Proj}_u v = \left(\frac{u\cdot v}{\|u\|^2}\right) u = \left(\frac{30}{49}\right)(3,-2,6)$$

$$= \left(\frac{90}{49}, -\frac{60}{49}, \frac{180}{49}\right)$$

(c) Compute $\mathbf{u} \times \mathbf{v}$ and then find the area of the parallellogram determined by $\mathbf{u}$ and $\mathbf{v}$.

$$u\times v = \begin{vmatrix} i & j & k \\ 3 & -2 & 6 \\ 4 & 0 & 3 \end{vmatrix} = i\begin{vmatrix} -2 & 6 \\ 0 & 3 \end{vmatrix} - j\begin{vmatrix} 3 & 6 \\ 4 & 3 \end{vmatrix} + k\begin{vmatrix} 3 & -2 \\ 4 & 0 \end{vmatrix}$$

$$= -6i + 15j + 8k$$

$\Rightarrow$ Area of parallelogram

$$= \|u\times v\| = \sqrt{(-6)^2+15^2+8^2} = \sqrt{325} = 5\sqrt{13}$$

Problem 2. (4 points)

(a) Find parametric equations for the line through the points $P(1,2,3)$ and $Q(6,3,5)$.

Line direction $V = \overrightarrow{PQ} = (5,1,2)$

Parametric equations:

$x = 1+5t, \quad y = 2+t, \quad z = 3+2t$

$-\infty < t < \infty$

(b) Find an equation for the plane through $P(0,2,-1)$ normal to the vector $3\mathbf{i} - 2\mathbf{j} - \mathbf{k}$.

$3(x-0) - 2(y-2) - (z-(-1)) = 0$

$3x - 2y + 4 - z - 1 = 0$

$3x - 2y - z = -3$